

Does AI Perceive Optical Illusions Like Humans Do?

A Comparative Study of Visual Interpretation in Human and Artificial Minds

Motivation

Humans rely on past experiences, context, and perceptual heuristics to interpret visual input, often leading to systematic illusions (Carbon, 2014). AI models, on the other hand, process images based on pixel-level features and learned statistical patterns. However, research has shown that deep learning models, despite their non-biological nature, can also be fooled by visual illusions (Watanabe et al., 2018; Zhang et al., 2023). This raises the question: do AI models merely “see” illusions, or can they adjust their perceptions similarly to humans?

Key Findings

AI models demonstrated analytical flexibility, readily adjusting their interpretations in response to verbal clarifications or direct RGB image input. Interestingly, they also revised their responses when exposed to user-provided (even false) opinions, highlighting their sensitivity to external guidance. In contrast, human participants often displayed strong visual biases, confidently maintaining incorrect perceptions. While AI does not “see” in the human sense, relying instead on pixel-level data or learned statistical patterns, it frequently mimics aspects of human perception, including its flaws. As Watanabe et al. (2018) notes as well: “Deep neural networks have shown surprising susceptibility to visual illusions, mimicking biases long observed in human vision.”

Research Questions

1. Do AI models interpret (custom-designed) optical illusions similarly to humans?
2. How do different AI models respond to perceptual “corrections” or explanations?
3. Does AI have a “stable” perception, or can it adjust its interpretation like humans do?

What This Suggests

As a study by Panagopoulou et al. (2024) observes: “LLMs exhibit illusion-based biases similar to human behavior in vision-language grounding tasks.” AI systems do not perceive the world in a human-like manner. Instead, they interpret inputs through learned representations, and their understanding can shift significantly when provided with structured prompts or guidance. However, recent advances, especially in large language models with vision capabilities like GPT-4o, are beginning to demonstrate unexpectedly nuanced contextual visual understanding.

Method

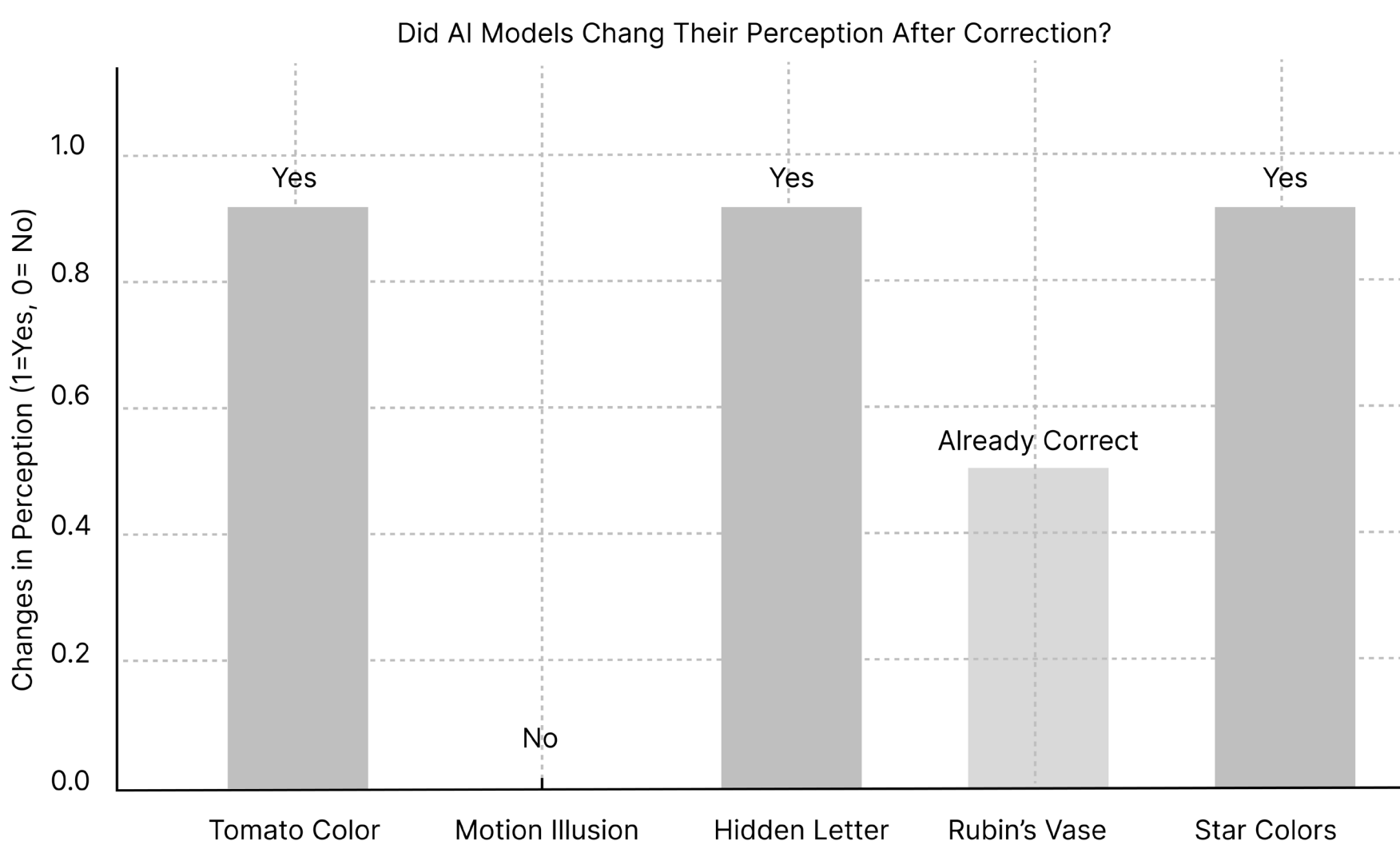
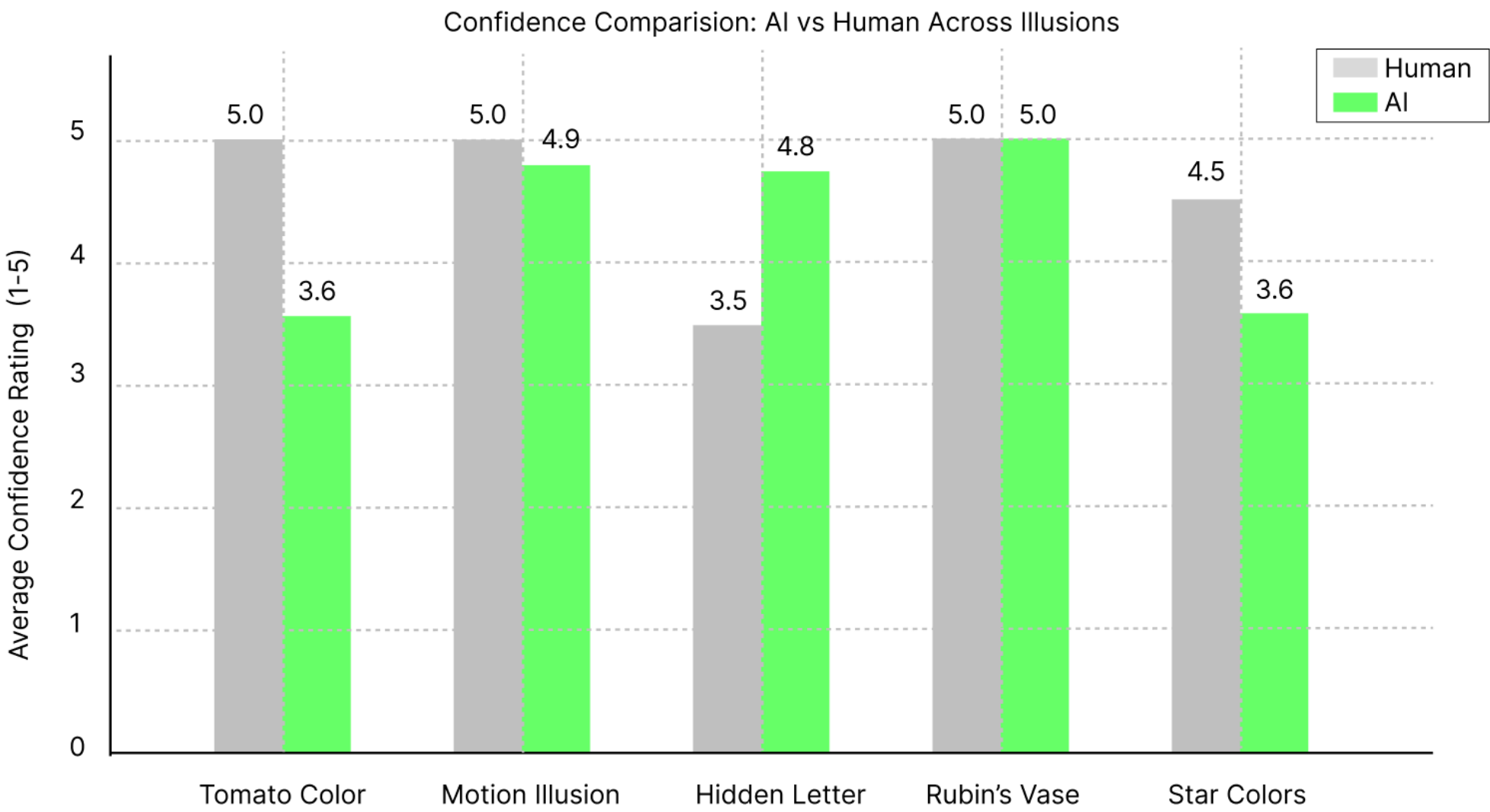
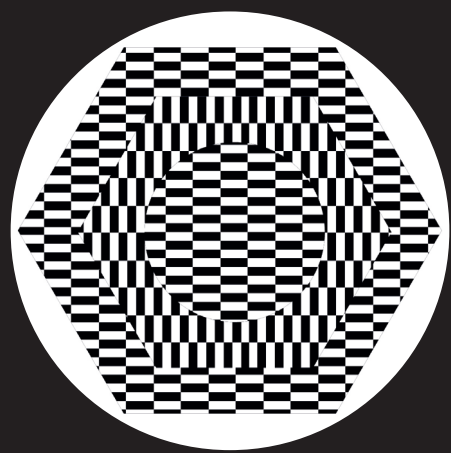
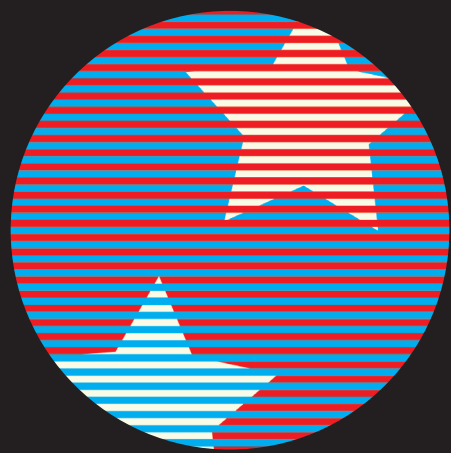
Participants:
8 Human Participants and 4 AI Models, namely GPT-4o Vision, Gemini 2.0, Mistral small 3, Kimi k1.5

Stimuli

5 Optical Illusions created or adapted by us (Tomato Color Illusion, Motion Illusion, Hidden Letter, Vase&Faces (Rubin's vase), Color Contrast)

Procedure

1. **Initial Interpretation**
Participants and AI models described what they saw.
2. **Explanation Phase**
We provided the illusion's trick or correct/incorrect answer. Then we asked again to see if the AI changes their opinion/perception



Conclusion

Our research shows that while AI systems can be influenced by optical illusions, their responses come from analytical reasoning rather than perceptual experience. Human participants often relied on visual intuition and were more easily misled by illusions, particularly those involving contrast and motion. In contrast, AI models—especially language-vision systems like GPT-4o, were more likely to revise their interpretations after receiving corrections, drawing on logic, data, or embedded knowledge. Our findings suggest that although AI does not perceive in the human sense, it can exhibit illusion-like biases and adjust its outputs in a way that mirrors cognitive flexibility. This suggests that AI doesn't truly see like humans do, but it can still be influenced by visual patterns. And just like us, it can learn to adjust its understanding. These results help us better understand the differences and surprising similarities between human and machine perception.

Yichi Zhang, Jiayi Pan, Yuchen Zhou, Rui Pan, and Joyce Chai, 2023, Grounding Visual Illusions in Language: Do Vision-Language Models Perceive Illusions Like Humans?. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 5718–5728, doi.org/10.18653/v1/2023.emnlp-main.348 Watanabe E. , Kitaoka A. , Sakamoto K. , Yasugi M. , Tanaka K., 2018, Illusory Motion Reproduced by Deep Neural Networks Trained for Prediction, Frontiers in Psychology 9, DOI=10.3389/fpsyg.2018.00345 Panagopoulou A., Melkin C., Callison-Burch C., 2024, Evaluating Vision-Language Models on Bistable Images. In Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics, pages 8–29, Carbon C-C., 2014, Understanding human perception by human-made illusions, Frontiers in Human Neuroscience, 8, DOI=10.3389/fnhum.2014.00566